

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459243.7534 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.046
Transit depth (Rp/Rs)^2	1.5 +/- 1.13 [%]
Orbital Inclination (inc)	85.96 +/- 1.81
Airmass coefficient 1 (a1)	0.998 +/- 0.039
Airmass coefficient 2 (a2)	0.003 +/- 0.023
Scatter in the residuals of the lightcurve fit is	3.88 %
Best Comparison Star	#1 - [269, 128]
Optimal Aperture	2.95
Optimal Annulus	6.56
Transit Duration (day)	0.027 +/- 0.047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.122 ± 0.046 vs 0.1295 (deviation 0.12σ)
Duration fit vs published	0.027 d vs 0.1146 d (deviation 1.43σ)
Mid-time O–C	-9.1 min
Depth SNR	2.0
Residual scatter	3.88 %

Light curve

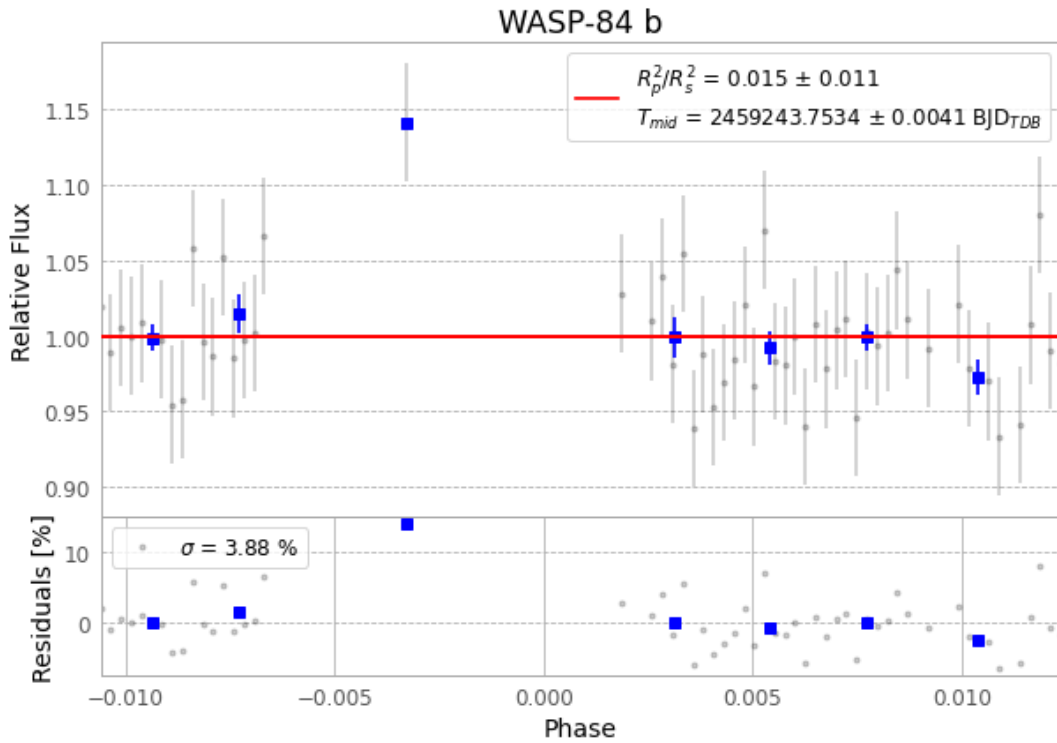


Figure 1 — FinalLightCurve_WASP-84 b_2021-01-28.png (SHA-256 cf61a1cfe3ff4aeb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b94e98bab912c2ac...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

95 FITS frames (62 MB), WASP-84210129034512.FITS → WASP-84210129082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-84-210129.log (ccde14464fcd...), FinalLightCurve_WASP-84 b_2021-01-28.png (cf61a1cfe3ff...), FinalLightCurve_WASP-84 b_2021-01-28.csv (ddddec3ebc6b...)

Compute resources

Wall time 355.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 07:25Z.